LIMESTONE AND KARST LANDFORMS / INDIA CAVES-MEGHALAYAN AGE

KARST COUPLED SYSTEM → CARBONATE CYCLE → KARST PREREQUISITES → HYDROLOGICAL ZONES → SURFACE-KARST LADDER → CAVE-DEVELOPMENT SEQUENCE

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

◆ PRIMARY CORE FLOW ◇ SUBORDINATE EXTRA ◆ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • active generation g3 • explicit approval of this exact generation is required • all prior artifacts unchanged

00

STAGE 00 KARST COUPLED SYSTEM

KARST COUPLED SYSTEM · JOINTS, BEDDING AND SINK POINTS · OPENINGS ENLARGE UNDERGROUND · SPRINGS EXPORT WATER AND SOLUTE · SURFACE + CAVE · ONE HYDROLOGICAL LANDSCAPE

RECHARGE → joints, bedding and sink points → DISSOLUTION → openings enlarge underground → CONDUITS → springs export water and solute → SURFACE + CAVE → one hydrological landscape

DEFINITION / COMPONENTS
- RECHARGE → joints, bedding and sink points
- DISSOLUTION → openings enlarge underground

TRIGGER / MECHANISM
- CONDUITS → springs export water and solute
- SURFACE + CAVE → one hydrological landscape

EFFECT / INTERVENTION
- RECHARGE → joints, bedding and sink points
- DISSOLUTION → openings enlarge underground

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CONDUITS → springs export water and solute; SURFACE + CAVE → one hydrological landscape.

01

STAGE 01 CARBONATE CYCLE

CARBONATE CYCLE · CO2 + WATER · WEAK CARBONIC ACID · ACID + CACO3 · DISSOLVED BICARBONATE · SOLUTION MOVES THROUGH CONDUITS · CALCITE CAN PRECIPITATE

CO2 + WATER → weak carbonic acid → ACID + CaCO3 → dissolved bicarbonate → TRANSPORT → solution moves through conduits → DEGASSING → calcite can precipitate

DEFINITION / COMPONENTS
- CO2 + WATER → weak carbonic acid
- ACID + CaCO3 → dissolved bicarbonate

TRIGGER / MECHANISM
- TRANSPORT → solution moves through conduits
- DEGASSING → calcite can precipitate

EFFECT / INTERVENTION
- CO2 + WATER → weak carbonic acid
- ACID + CaCO3 → dissolved bicarbonate

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRANSPORT → solution moves through conduits; DEGASSING → calcite can precipitate.

02

STAGE 02 KARST PREREQUISITES

KARST PREREQUISITES · SOLUBLE, THICK AND EXPOSED TO RECHARGE · JOINTS AND BEDDING PATHWAYS · GRADIENT PLUS OUTLET · BASE LEVEL · SUSTAINED DISSOLUTION AND REMOVAL

ROCK → soluble, thick and exposed to recharge

STRUCTURE → joints and bedding pathways → FLOW → gradient plus outlet / base level → TIME → sustained dissolution and removal

DEFINITION / COMPONENTS
- ROCK → soluble, thick and exposed to recharge
- STRUCTURE → joints and bedding pathways

TRIGGER / MECHANISM
- FLOW → gradient plus outlet / base level
- TIME → sustained dissolution and removal

EFFECT / INTERVENTION
- ROCK → soluble, thick and exposed to recharge
- STRUCTURE → joints and bedding pathways

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FLOW → gradient plus outlet / base level.

03

STAGE 03 HYDROLOGICAL ZONES

HYDROLOGICAL ZONES · SHALLOW STORAGE AND ROUTING · DESCENDING WATER IN AIR-FILLED VOIDS · WATER TABLE · FLUCTUATING BOUNDARY · SATURATED TUBES AND LOOPS

EPKARST → shallow storage and routing

VADOSE → descending water in air-filled voids → WATER TABLE → fluctuating boundary → PHREATIC → saturated tubes and loops

SYSTEM / DEFINITION	PROCESS / MECHANISM	SPATIAL / INDIA PATTERN	HAZARD / LIMIT / RESPONSE
• EPKARST → shallow storage and routing	• WATER TABLE → fluctuating boundary	• EPKARST → shallow storage and routing	• WATER TABLE → fluctuating boundary
• VADOSE → descending water in air-filled voids	• PHREATIC → saturated tubes and loops	• VADOSE → descending water in air-filled voids	• PHREATIC → saturated tubes and loops

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WATER TABLE → fluctuating boundary; PHREATIC → saturated tubes and loops.

04

STAGE 04 SURFACE-KARST LADDER

SURFACE-KARST LADDER · SMALL SOLUTION GROOVES · CLOSED DEPRESSION · LARGER COMPOSITE · STRUCTURAL BASIN · SWALLOW HOLE · SURFACE STREAM TRANSFERS UNDERGROUND

KARREN → small solution grooves

DOLINE → closed depression → UVALA / POLJE → larger composite / structural basin → SWALLOW HOLE → surface stream transfers underground

DEFINITION / COMPONENTS
- KARREN → small solution grooves
- DOLINE → closed depression

TRIGGER / MECHANISM
- UVALA / POLJE → larger composite / structural basin
- SWALLOW HOLE → surface stream transfers underground

EFFECT / INTERVENTION
- KARREN → small solution grooves
- DOLINE → closed depression

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: UVALA / POLJE → larger composite / structural basin.

05

STAGE 05 CAVE-DEVELOPMENT SEQUENCE

CAVE-DEVELOPMENT SEQUENCE · FAVOURABLE FRACTURE OR BEDDING PLANE · PASSAGE ENLARGES UNDER SATURATION · STREAM INCISES AFTER WATER-TABLE FALL · LATE CHANGE · COLLAPSE, FILL OR RENEWED SOLUTION

INCEPTION → favourable fracture or bedding plane → PHREATIC → passage enlarges under saturation → VADOSE → stream incises after water-table fall → LATE CHANGE → collapse, fill or renewed solution

DEFINITION / COMPONENTS
- INCEPTION → favourable fracture or bedding plane
- PHREATIC → passage enlarges under saturation

TRIGGER / MECHANISM
- VADOSE → stream incises after water-table fall
- LATE CHANGE → collapse, fill or renewed solution

EFFECT / INTERVENTION
- INCEPTION → favourable fracture or bedding plane
- PHREATIC → passage enlarges under saturation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LATE CHANGE → collapse, fill or renewed solution.

06

STAGE 06 SPELEOTHEM MAP

SPELEOTHEM MAP · ROOF DRIP DEPOSIT · FLOOR SPLASH · DRIP DEPOSIT · ROOF AND FLOOR FORMS JOIN · FILM FLOW ON WALL OR SLOPE

SYSTEM / DEFINITION	PROCESS / MECHANISM	SPATIAL / INDIA PATTERN	HAZARD / LIMIT / RESPONSE
STALACTITE → roof drip deposit	COLUMN → roof and floor forms join	STALACTITE → roof drip deposit	COLUMN → roof and floor forms join
STALAGMITE → floor splash / drip deposit	FLOWSTONE / CURTAIN → film flow on wall or slope	STALAGMITE → floor splash / drip deposit	FLOWSTONE / CURTAIN → film flow on wall or slope

SYSTEM / DEFINITION
- STALACTITE → roof drip deposit
- STALAGMITE → floor splash / drip deposit

PROCESS / MECHANISM
- COLUMN → roof and floor forms join
- FLOWSTONE / CURTAIN → film flow on wall or slope

SPATIAL / INDIA PATTERN
- STALACTITE → roof drip deposit
- STALAGMITE → floor splash / drip deposit

HAZARD / LIMIT / RESPONSE
- COLUMN → roof and floor forms join
- FLOWSTONE / CURTAIN → film flow on wall or slope

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FLOWSTONE / CURTAIN → film flow on wall or slope.

07

STAGE 07 INDIA CAVE FIREWALL

INDIA CAVE FIREWALL · MEGHALAYA LIMESTONE · NATURAL CAVES IN ANDHRA PRADESH · HUMAN ROCK-CUT ARCHITECTURE · LOCATION DOES NOT REPLACE GENETIC CLASSIFICATION

SYSTEM / DEFINITION	PROCESS / MECHANISM	SPATIAL / INDIA PATTERN	HAZARD / LIMIT / RESPONSE
MAWMLUH / MAWSMAI / SIJU → Meghalaya limestone	AJANTA / ELLORA → human rock-cut architecture	MAWMLUH / MAWSMAI / SIJU → Meghalaya limestone	AJANTA / ELLORA → human rock-cut architecture
BORRA / BELUM → natural caves in Andhra Pradesh	RULE → location does not replace genetic classification	BORRA / BELUM → natural caves in Andhra Pradesh	RULE → location does not replace genetic classification

SYSTEM / DEFINITION
- MAWMLUH / MAWSMAI / SIJU → Meghalaya limestone
- BORRA / BELUM → natural caves in Andhra Pradesh

PROCESS / MECHANISM
- AJANTA / ELLORA → human rock-cut architecture
- RULE → location does not replace genetic classification

SPATIAL / INDIA PATTERN
- MAWMLUH / MAWSMAI / SIJU → Meghalaya limestone
- BORRA / BELUM → natural caves in Andhra Pradesh

HAZARD / LIMIT / RESPONSE
- AJANTA / ELLORA → human rock-cut architecture
- RULE → location does not replace genetic classification

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → location does not replace genetic classification.

08

STAGE 08 PROXY PIPELINE

PROXY PIPELINE · RAINFALL, VEGETATION AND SOIL CO2 · ROUTE AND RESIDENCE TIME · ELEMENT SIGNAL · DATED SERIES WITH UNCERTAINTY

CLIMATE → rainfall, vegetation and soil CO2 → RECHARGE → route and residence time → CALCITE → isotope / element signal → CHRONOLOGY → dated series with uncertainty

DEFINITION / COMPONENTS
- CLIMATE → rainfall, vegetation and soil CO2
- RECHARGE → route and residence time

TRIGGER / MECHANISM
- CALCITE → isotope / element signal
- CHRONOLOGY → dated series with uncertainty

EFFECT / INTERVENTION
- CLIMATE → rainfall, vegetation and soil CO2
- RECHARGE → route and residence time

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CALCITE → isotope / element signal; CHRONOLOGY → dated series with uncertainty.

09

STAGE 09 MEGHALAYAN HIERARCHY

MEGHALAYAN HIERARCHY · KM-A SPELEOTHEM AT MAWMLUH · QUATERNARY → PERIOD · HOLOCENE → EPOCH

QUATERNARY → Period

HOLOCENE → Epoch → MEGHALAYAN → Age / Stage → GSSP → KM-A speleothem at Mawmluh

DEFINITION / COMPONENTS
- QUATERNARY → Period
- HOLOCENE → Epoch

TRIGGER / MECHANISM
- MEGHALAYAN → Age / Stage
- GSSP → KM-A speleothem at Mawmluh

EFFECT / INTERVENTION
- QUATERNARY → Period
- HOLOCENE → Epoch

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MEGHALAYAN → Age / Stage; GSSP → KM-A speleothem at Mawmluh.

10

STAGE 10 STATUS AND NARRATIVE CAUTION

STATUS AND NARRATIVE CAUTION · 4.2 KA EVENT · CORRELATION GUIDE · GSSP POINT · PHYSICAL LOWER BOUNDARY · NAMED FORMAL INTERVAL · NOT A RATIFIED EPOCH

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
4.2 ka EVENT → correlation guide	MEGHALAYAN → named formal interval	4.2 ka EVENT → correlation guide
GSSP POINT → physical lower boundary	ANTHROPOCENE → not a ratified Epoch	GSSP POINT → physical lower boundary

DEFINITION / COMPONENTS
- 4.2 ka EVENT → correlation guide
- GSSP POINT → physical lower boundary

TRIGGER / MECHANISM
- MEGHALAYAN → named formal interval
- ANTHROPOCENE → not a ratified Epoch

EFFECT / INTERVENTION
- 4.2 ka EVENT → correlation guide
- GSSP POINT → physical lower boundary

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MEGHALAYAN → named formal interval; ANTHROPOCENE → not a ratified Epoch.

11

STAGE 11 KARST CONSERVATION SPINE

KARST CONSERVATION SPINE · RECHARGE, PASSAGES, SPRINGS AND VALUES · IRREVERSIBLE QUARRY AND POLLUTION PATHWAYS · VISITATION, WASTE AND WATER USE · HYDROLOGY, BIODIVERSITY AND STRUCTURAL CHANGE

MAP → recharge, passages, springs and values → AVOID → irreversible quarry and pollution pathways → MANAGE → visitation, waste and water use → MONITOR → hydrology, biodiversity and structural change

DEFINITION / COMPONENTS
- MAP → recharge, passages, springs and values
- AVOID → irreversible quarry and pollution pathways

TRIGGER / MECHANISM
- MANAGE → visitation, waste and water use
- MONITOR → hydrology, biodiversity and structural change

EFFECT / INTERVENTION
- MAP → recharge, passages, springs and values
- AVOID → irreversible quarry and pollution pathways

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MANAGE → visitation, waste and water use; MONITOR → hydrology, biodiversity and structural change.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · MEGHALAYA HAS INDIA'S PREMIER KARST SYSTEMS. KREM LIAT PRAH · THE CHERRAPUNJI LIMESTONE CAVE SHOWS WELL-DEVELOPED STALACTITE AND STALAGMITE · CAVE TOURISM SPANS TWO DIFFERENT CATEGORIES · NATURAL KARST CAVES IN MEGHALAYA AND ROCK-CUT/ · VERY HIGH RAINFALL OVER THICK LIMESTONE MAKES MEGHALAYA IDEAL · THESE CAVES DOUBLE AS PALAEOCLIMATE ARCHIVES — THEIR STALAGMITES · GLOBAL BOUNDARY STRATOTYPE SECTION AND POINT — THE PHYSICAL

OPTIONAL PROCESS DEPTH
- Meghalaya has India's premier karst systems. Krem Liat Prah is widely cited as India's longest
- The Cherrapunji limestone cave shows well-developed stalactite and stalagmite formations (dripstone deposition).
- Cave tourism spans two different categories: natural karst caves in Meghalaya and rock-cut/
- Very high rainfall over thick limestone makes Meghalaya ideal for rapid solution and cave growth.

DATA / INSTITUTION LIMIT
- These caves double as palaeoclimate archives — their stalagmites record past rainfall/drought.
- GSSP = Global Boundary Stratotype Section and Point — the physical 'golden spike' defining a time boundary.
- The Meghalayan Age is the youngest stage of the Holocene (last ~4,200 years); its GSSP is the KNI-51 stalagmite in Mawmluh Cave, Meghalaya.
- The stalagmite isotope signal marks abrupt weakening of the Indian summer monsoon around 4.2 ka.

CONTEMPORARY PARALLEL
- Krem Liat Prah in Meghalaya is widely cited as India's longest natural cave; mapped length is survey-dependent.
- Cherrapunji limestone cave: developed stalactite/stalagmite (Khullar).
- Ajanta (Maharashtra), Pachmarhi (MP ~500 caves), Khandagiri/Udayagiri (Odisha).
- High rainfall + thick limestone = rapid karst solution in Meghalaya.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels